

DEPENDABILITY AND RELIABILITY OF MANUFACTURING SYSTEMS

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Abstract: *Manufacturing systems are characterized by three fundamental properties: performance, cost, and dependability. The analysis of a system from the pure performance viewpoint tends to be optimistic since it ignores the failure/repair behaviour in the system. On the other hand, pure availability analysis tends to be too conservative since performance considerations are not taken into account.*

They improve product quality and reduce production losses. Dependability of a manufacturing system is the ability to deliver service (products) that can justifiably be trusted. Performance engineering addresses sustainability along with other factors like quality, reliability, maintainability and safety.

The current paper introduces a framework for the fault analysis of production process, and the availability evaluation especially in the Flexible Manufacturing Systems (FMS's). Generally, computing dependability measures of repairable systems with usual failure and repair processes is difficult, through either analytical or numerical methods. Mathematical models and application software for this kind of problems are in good stead. Recent advances in neural networks show that they can be used in applications that involve predictions, including the manufacturing.

Key words: dependability, reliability, manufacturing systems

1. INTRODUCTION

Plant Wellness is a systematic method, with the availability as kernel, for maximum operating profit (maximizing potential) through failure elimination (disease-free), risk management and proactive maintenance (seeking complete well-being), Lean waste removal (integrated and dynamic performance) and idle quality assurance (self-responsibility) [19].

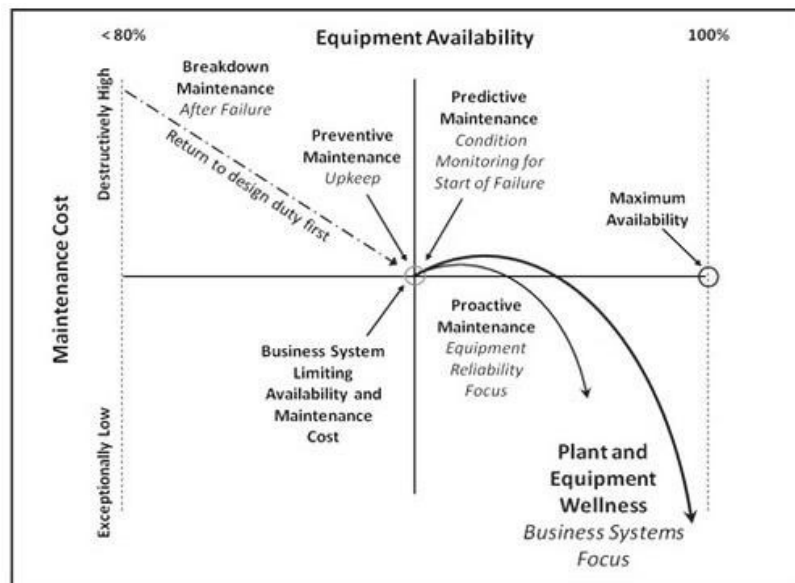


Fig. 1. The plant wellness way bring low maintenance costs by creating high reliability assets [19]

Plant Wellness is a perfect metaphor for industrial maintenance and asset management success. Putting the write maintenance processes and reliability practices into the maintenance and reliability system will fix the reliability and maintenance problems [19].

Reliability of manufacturing equipment has a large impact on throughput and productivity. Hitherto, attempts at modeling manufacturing equipment reliability has concentrated on the use of analytical models or high-level simulation models [3]. At the same time it is important to have available different stastical methods for practical use. Focus must be on reliability and not cost, because if reliability starts to improve the cost will definitely go down and it cannot be the other way around [19].

2. DEPENDABILITY AND PERFORMABILITY

The service delivered by a system is its behavior as it is perceived by its user(s), which is another system (physical, human) that interacts with the former at the service interface [2]. Based on the definition of the failure, an alternate definition of dependability emerges, which complements the initial definition in providing a criterion for establishing whether the delivered service can be trusted or not: the ability of a system to avoid failures that are more frequent or more severe, and outage durations that are longer, than is acceptable to the user(s). In the opposite case, the system is no longer dependable: it suffers from a dependability

failure that is a meta-failure. In standards it is possible to appear many examples of reliability, maintainability, the logistical support of maintenance and availability requirements [2].

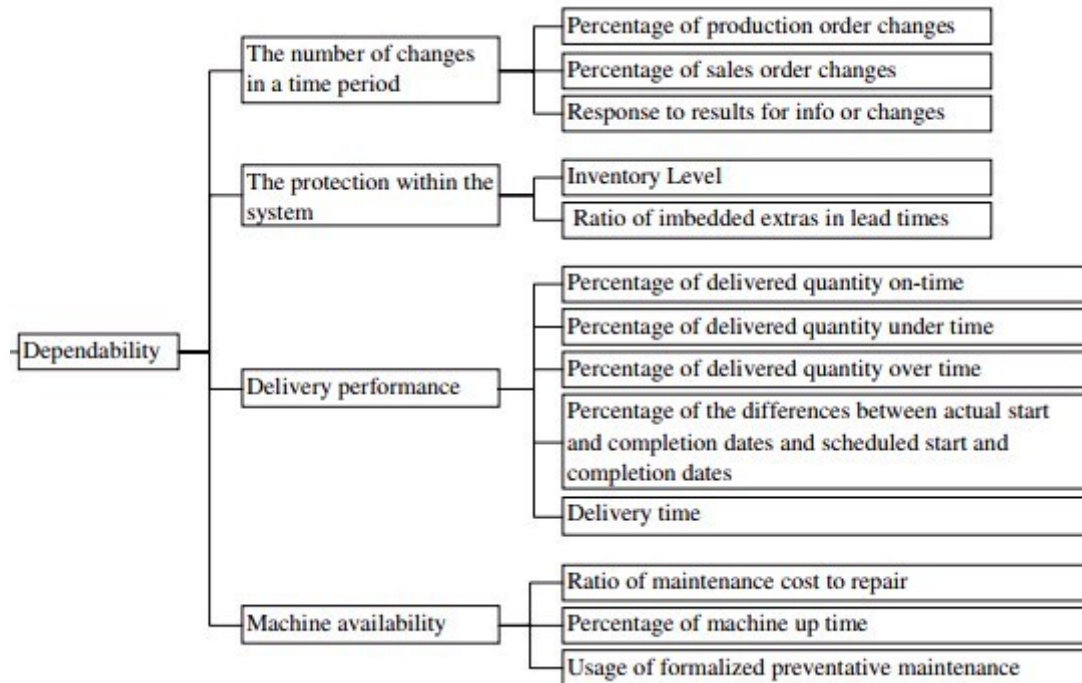


Fig.2. The hierarchical framework of the manufacturing system's performance evaluation modified from Yurdakul [17].

If the dependability of a manufacturing system increases, there would be less dependence on unexpected and expected deviations and delays to protect the system. For dependability, some of the performance measures are the number of changes in time periods, performance delivery, and machine availability (Fig. 2) [5].

Generally, individual assessments of system performance and dependability are not so easily combined, particularly if performance in the presence of faults is degradable, i.e., fault-caused errors can reduce the quality of a delivered service even though that service, according to its specification, remains proper (failure-free) [7], [10]. The need to accommodate this property, using model-based evaluation methods, was the *raison d'être* for the concept of performability, introduced by John Meyer [7].

Performance engineering can be defined as the entire engineering effort to improve the performance of a system to ensure high quality, reliability, maintainability and safety and is also sustainable [8]. Important is the high performance of a system and its minimum lifecycle costs and improved performance should necessarily imply less environmental achieving sustainability for a dependable product, called performability engineering; it reflects the entire engineering effort of a producer to achieve the performability of a product, system or service,

which in fact can be called improving 3-S: survivability, safety and sustainability (fig. 3) [8]. The usual definition of dependability ignores the accompanying environmental consequences while creating products, systems and services [9]. Performability takes a holistic view of various activities and processes and takes stock of what is being produced and what [16]. Significant results in performability evaluation of manufacturing systems producing single part type have been obtained by Viswanadham [15].

3. MANUFACTURING APPLICATIONS

A manufacturing system includes a set of machines performing different operations, linked by a material handling system (MHS). A major consideration in designing a manufacturing system is its performance. The models existing in the literature do not consider simultaneous performance of various manufacturing hardware components of the system.



Fig.4. Flexible Manufacturing Cell [1]

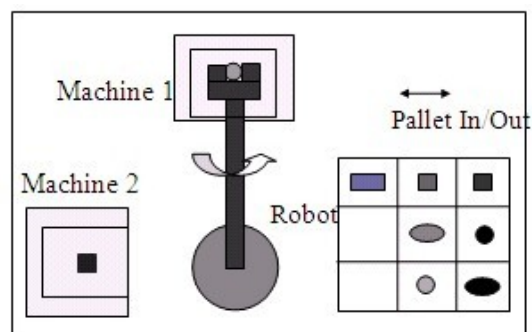


Fig. 3 Performability [8]

When a machine or any other hardware component of the system fails, the system reconfiguration is often less than perfect. The notion of imperfection is called imperfect coverage, and it is defined as probability that the system successfully reconfigures given that component faults occur [18]. A performance evaluation for manufacturing equipment is the technological reliability, as the operating time within the required inaccuracy, that depends on the condition of the machine accuracy (i.e. how well the machine-tool was originally set) and the speed of evolution of each parameter accuracy (generally affected by wear of bearings and machine-tool slideways), however, assuming that the machine-tools were properly adjusted, a decisive test of accuracy will be considered the one with the highest speed of development. The case study is referring to technological reliability $R_{th}(t)$ of a universal milling machine with a high level of geometric accuracy parameters [12].

The criterion of the time variation of the geometric accuracy was selected for the establishing of technological reliability indicators. It was monitored a set of universal milling machines in the factories working conditions for a time span for the reliability evaluations. A set of measurements was made for the accuracy of geometry [12].

In a practical case, approximating regression functions by straight regression, the rate of evolution is proportional to the slope of the straight line [11]. The reliability community has become well experienced in fitting of survival distributions, the use of design of experiments (DOE) and the associated general linear model (linear regression and analysis of variance methods) approach to analysis [11].

Different configurations of the Flexible Manufacturing Cells are represented in fig.4 and 5. Fig.5 presents the Markov model image for a manufacturing cell (Fig.4), as basis for dependability evaluation [14]. The states of a manufacturing cell are analyzed from the functional point of view. A new model has been conceived based on the graph Markov properties, and residue classes, which further offers the possibility to determine the system availability and productivity [14].

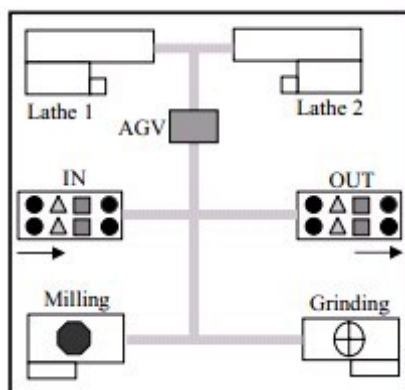


Fig.5. Flexible Manufacturing

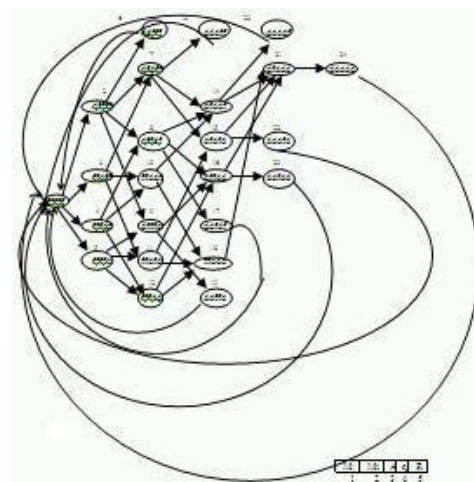


Fig. 6. Flexible Manufacturing Cell with AGV [15] Cell with AGV [13]

4. SOFTWARE APPLICATIONS

In real systems, availability, and performance are important quality of service (QoS) indices which should be studied in a composite manner. The combined evaluation of the indices described above is useful when the system under study can operate in a gracefully degradable

manner in the presence of component failures. SHARPE and SPNP software packages that were developed allow such combined performance/availability analysis [4]. SHARPE, (Symbolic Hierarchical Automated Reliability and Performance Evaluator) is a tool for specifying and analyzing performance, reliability and performability models. It has been installed at over 250 sites. Model types include combinatorial one such as fault-trees and queuing networks and state-space ones such as Markov and semi-Markov reward models as well stochastic Petri nets. Steady-state, transient and interval measures can be computed. Output measures of a model can be used as parameters of other models. This facilitates the hierarchical combination of different model types [4]. SPNP (Stochastic Petri Net Package) used for input is a stochastic reward net (**SRN**). SRNs incorporate several structural extensions to **GSPNs** such as marking dependencies (marking dependent arc cardinalities, guards, etc.) and allow reward rates to be associated with each marking. The reward function can be marking dependent as well. They are specified using **CSPL** (C based SRN Language) which is an extension of the C programming language with additional constructs for describing the SRN models. SRN specifications are automatically converted into a Markov reward model which is then solved to compute a variety of transient, steadystate, cumulative, and sensitivity measures. For SRNs with absorbing markings, mean time to absorption and expected accumulated reward until absorption can be computed [4].

Neural-network models have a significant advantage over analytic models, though, because they require only failure history as input, no assumptions. Using that input, the neuralnetwork model automatically develops its own internal model of the failure process and predicts future failures. Because it adjusts model complexity to match the complexity of the failure history, it can be more accurate than some commonly used analytic models [6]. Practically, after training the neural network with a failure history up to time t (where t is less than the total testing and debugging time of 44 days), you can use the network to predict the cumulative faults at the end of a future testing and debugging session [6].

5. CONCLUSIONS

Dependability and reliability of production processes are key issues for the stability of production system operation. Dependability measures a company's planning and control system's effectiveness and determines the responsive level in a manufacturing system. In a dependable manufacturing system, all ordered products should be produced and delivered in time and in ordered quantities. At the same time, dependability has been used to include more attributes related to performance. Therefore, it was considered logical and appropriate to extend the meaning of performability to include attributes like dependability and sustainability. It cannot separate environmental problems from the economics of clean production and clean technologies. The definition of the term performability has been widened to include sustainability in the context of the changed scenario of the 21st century in order to reflect a holistic view of designing, producing and using products, systems or services, which will satisfy the performance requirements of a customer to the best possible extent and are not only dependable (implying survivability and safety) but are also sustainable. Software toolkits provide a specification language and solution methods for most of the commonly used model types for performance, reliability and performability modeling. Recent advances in neural networks show that they can be used in applications that involve predictions. An interesting and difficult application is time-series prediction, which predicts a complex sequential process like reliability growth. One drawback of neural networks is that you can't interpret the knowledge stored in their weights in simple terms that are directly related to software metrics -which is something you can do with some analytic models.

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